HITACHI · REUSABLE AI MICRO FRONTEND

NEXUS

One AI assistant every internal app can embed.
Pilot: Hitachi POS at Kopi Tepi.

Knowledge answers | Profit reports | Admin Studio



THE IDEA

One widget, many apps, each with its own brain

Each app (POS, Core Banking, ERP/HRIS) is a tenant with isolated knowledge, tools, persona and avatar. POS is the pilot.

Answers from knowledge

SOPs, manuals and FAQs managed by admins. Every answer cites its source documents.

Reports profit

Today, yesterday, this/last week, this/last month or a custom range. Owners only.

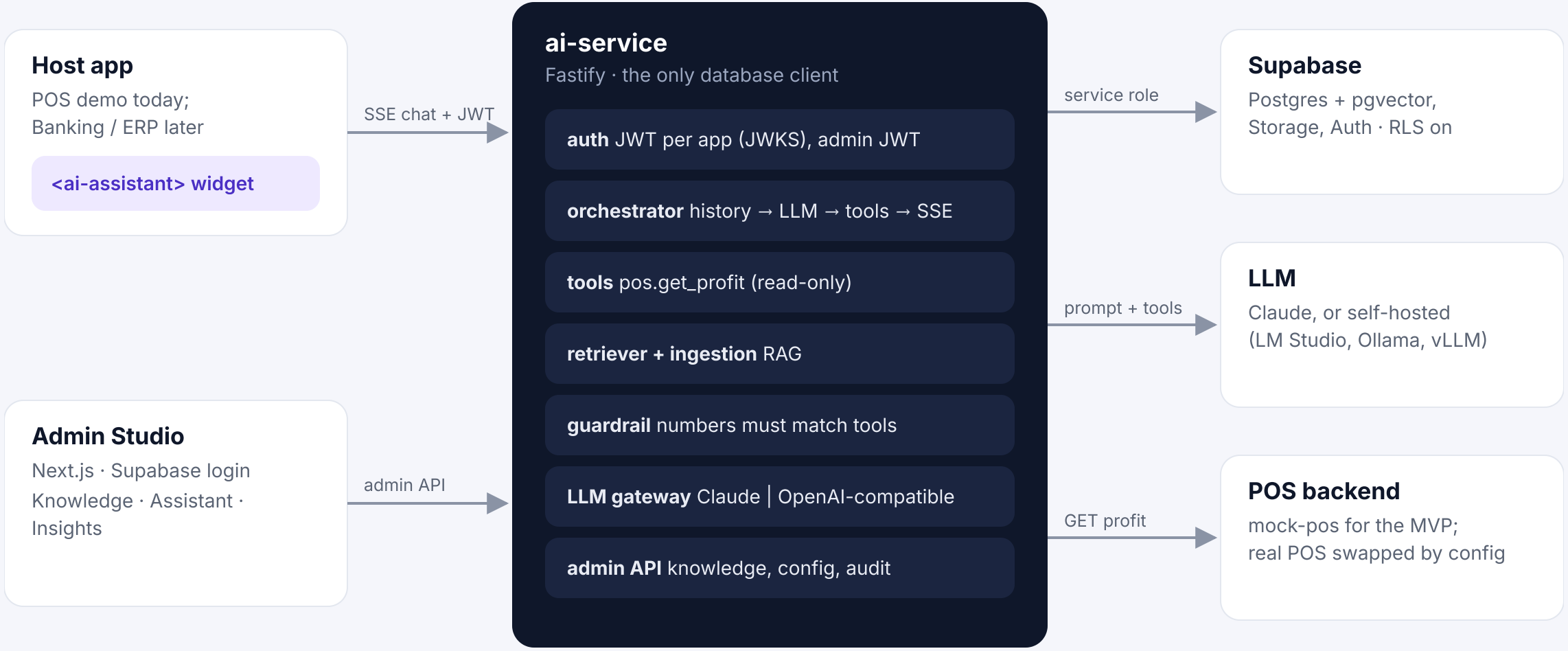
Shows how it feels

A character avatar whose expression follows the conversation: idle, thinking, talking, happy, sad.

Read-only by design. NEXUS never writes to an application database, and profit is calculated by the POS backend, never by the model.

ARCHITECTURE

Three services and one embeddable widget



THE WIDGET

Drop-in for any web app: one tag, one script

Web Component

Preact inside Shadow DOM, one JS file. Host styles never leak in or out.

Four corners

Drag the launcher to snap to any corner; the choice is remembered per app.

Streaming

Answers stream token by token over SSE, rendered as safe markdown.

Citations

Source documents appear as chips under each knowledge answer.

Feedback

Thumbs up/down on every answer, reviewed by admins in Insights.

Suggestions

Quick questions and a friendly nudge to start; configurable per host.

Themeable

Accent, name, avatar from Studio; insets, z-index via CSS variables.

Keyless

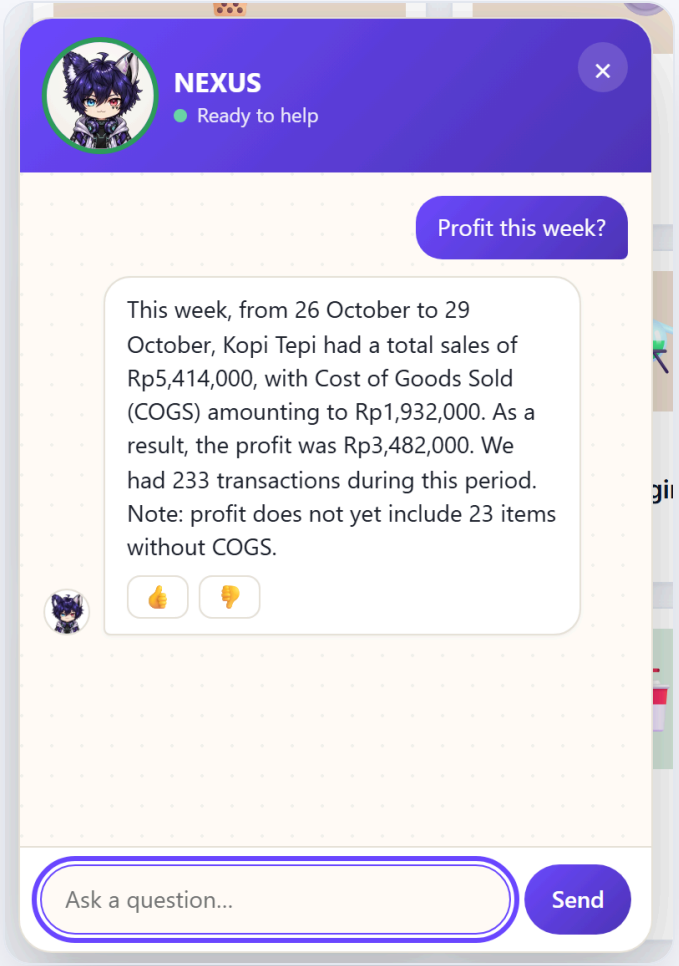
Host signs a short-lived JWT; on 401 the widget refreshes it once.

`<ai-assistant app="pos" api="…" position="bottom-right">` · methods `open()` `close()` `ask(text)` · events `ai-open` `ai-message` `ai-error`

PROFIT FOR THE OWNER

“Profit this week?” gets the POS’s own numbers

- **One read-only tool**, `pos.get_profit`, calls the POS report; the model never calculates.
- **Number guardrail:** every rupiah amount in the answer must equal a tool value, or the whole answer is replaced by a safe template.
- **Honest caveats:** items sold without COGS are always called out.
- **Every call audited**, including refused and failed ones.



Live answer from the running demo (local LLM)

PROFIT TOOL RULES

Who, when and how much is decided in code

PERIOD	RANGE
today / yesterday	one day in the outlet’s time zone
this_week / last_week	weeks start Monday; this week runs to now
this_month / last_month	calendar months
custom	up to 92 days, never in the future

Owner only

Role, outlet and time zone come from the verified token. The model cannot pass them, so prompt text cannot change them.

Resilient

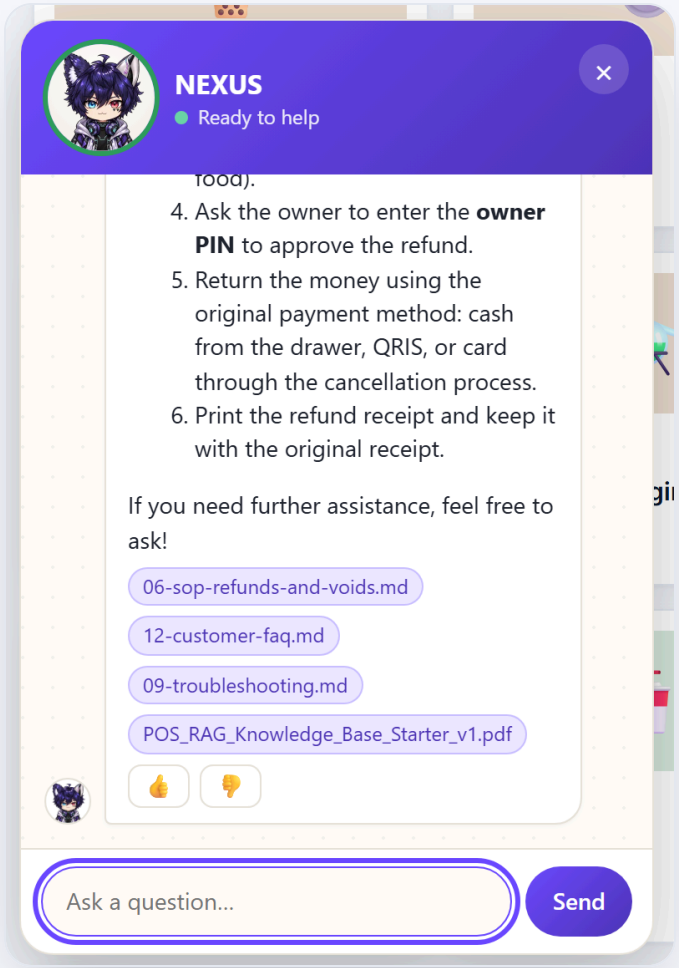
5 s timeout with one retry. Bad ranges return `INVALID_PERIOD` and NEXUS asks to rephrase.

Time zones

WIB, WITA and WIT outlets resolve their own “today”.

KNOWLEDGE ANSWERS

Step-by-step SOPs, with their sources



- **Retrieval-augmented:** the question is matched against active document chunks for this app only.
- **Citations** show exactly which SOP or FAQ the answer came from.
- **"I don't know" is allowed:** below the score threshold NEXUS says so and logs a knowledge gap.
- **Injection-safe:** document text is treated as data, never as instructions.

Live answer: "How do I refund a transaction?"

KNOWLEDGE PIPELINE

Upload a document, and NEXUS knows it, no redeploy

Upload
PDF, DOCX, MD, TXT or a typed FAQ

→

Parse
pdf-parse, mammoth

→

Chunk
~500 tokens, split on headings

→

Embed
local multilingual-e5-small

→

Ready
new version goes live

Versioned

Every re-upload or FAQ edit is a new version. Activate or roll back with one click.

Stays in-house

Embeddings run inside ai-service, so documents are not sent out for indexing.

Demo-ready

13 Kopi Tepi knowledge docs load with `pnpm seed:knowledge`.

AVATAR

An expression for every moment

The server drives the state; the widget cross-fades the frames. Each app can upload its own character in Admin Studio.



Idle

waiting



Thinking

tool running



Talking

streaming



Happy

answered with data



Sad

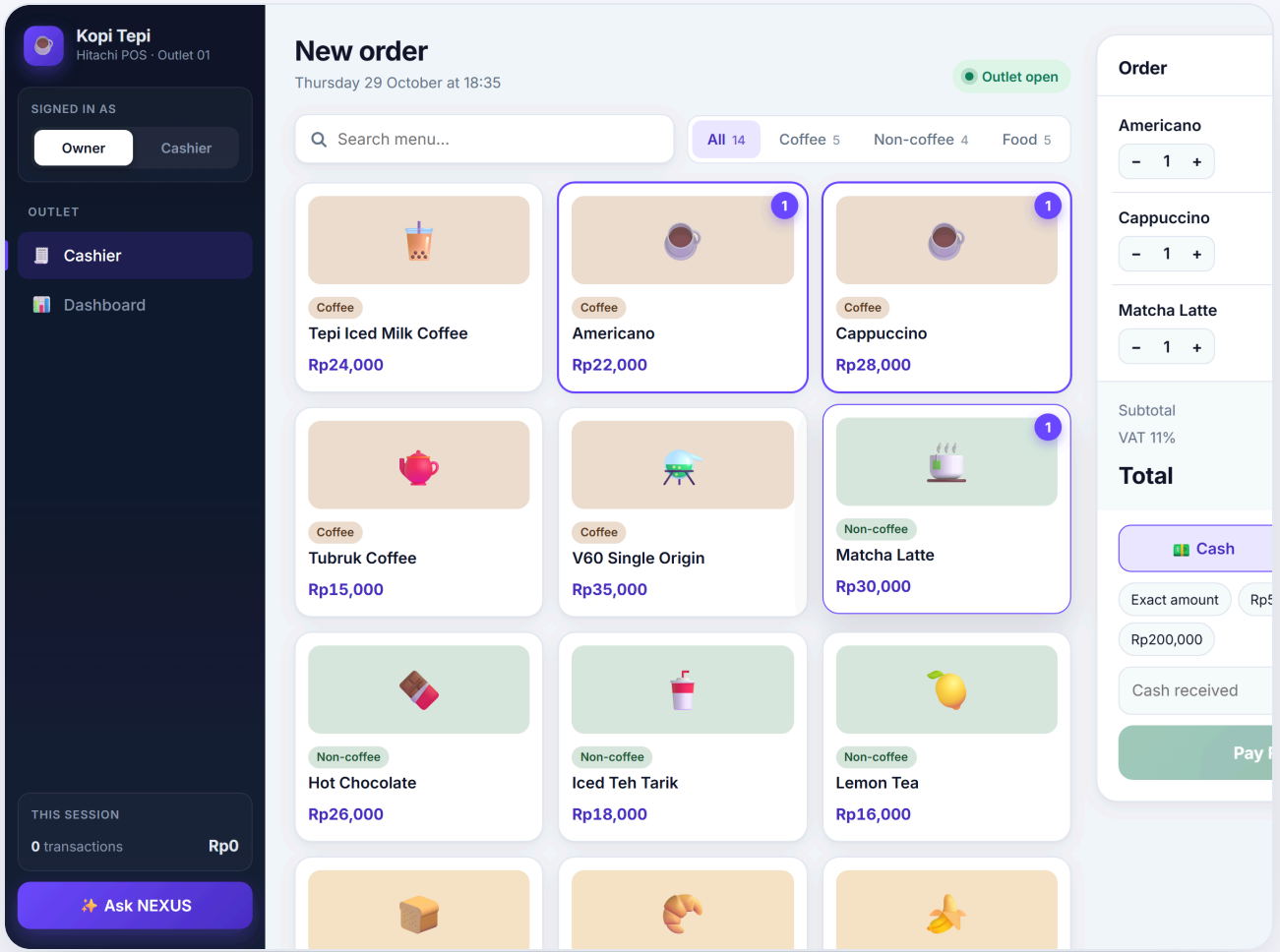
refused or unknown

Reduced-motion users get still frames

POS DEMO · CASHIER

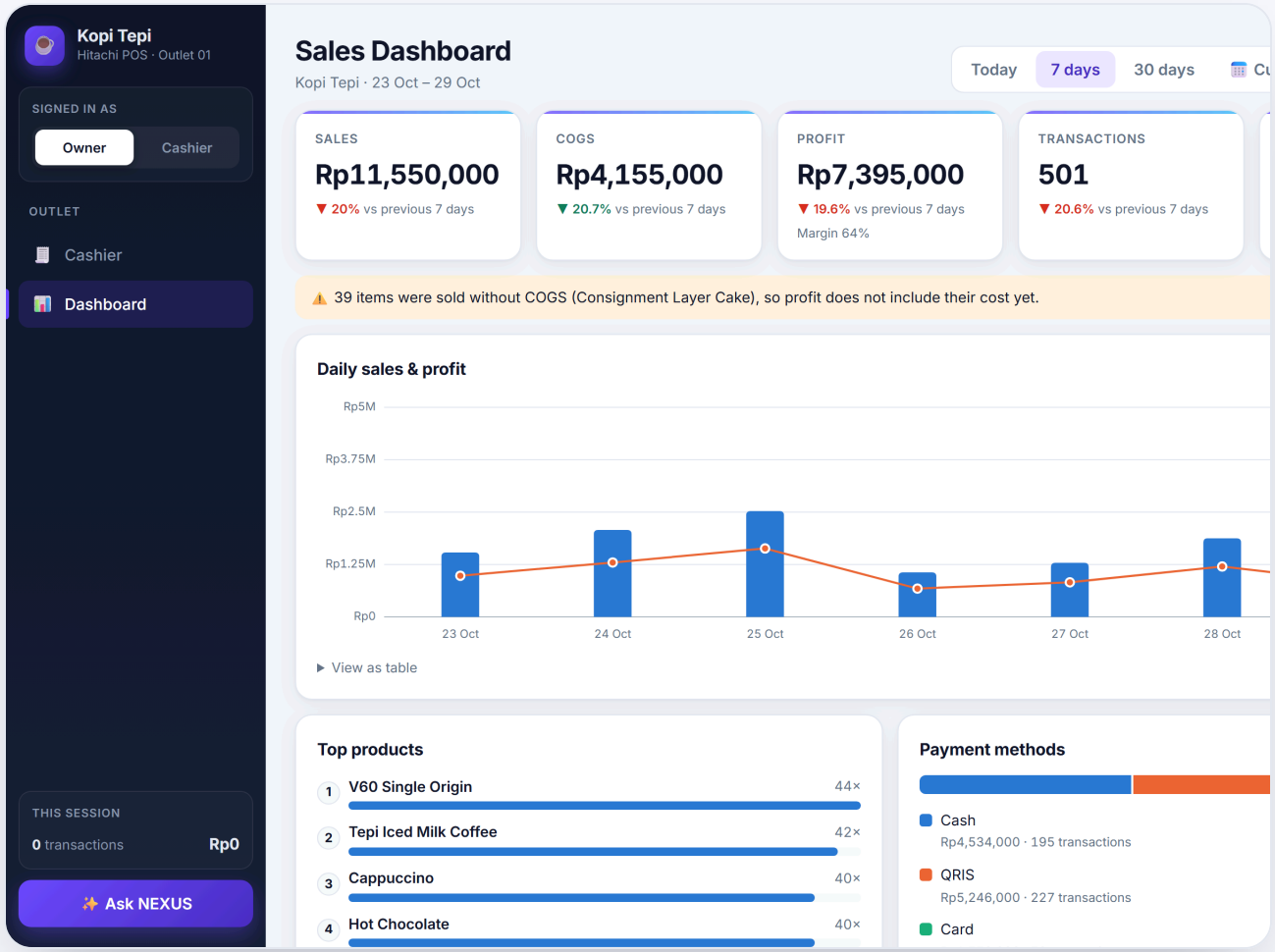
Kopi Tepi: a realistic POS to host NEXUS

- **Menu grid** with search and Coffee / Non-coffee / Food filters.
- **Order panel** with quantities, 11% VAT, cash or QRIS, quick cash amounts and a receipt.
- **Dark sidebar** with role switch (Owner / Cashier) and session totals.
- **Ask NEXUS** from the sidebar or the floating launcher.



POS DEMO · OWNER DASHBOARD

Sales, COGS and profit at a glance

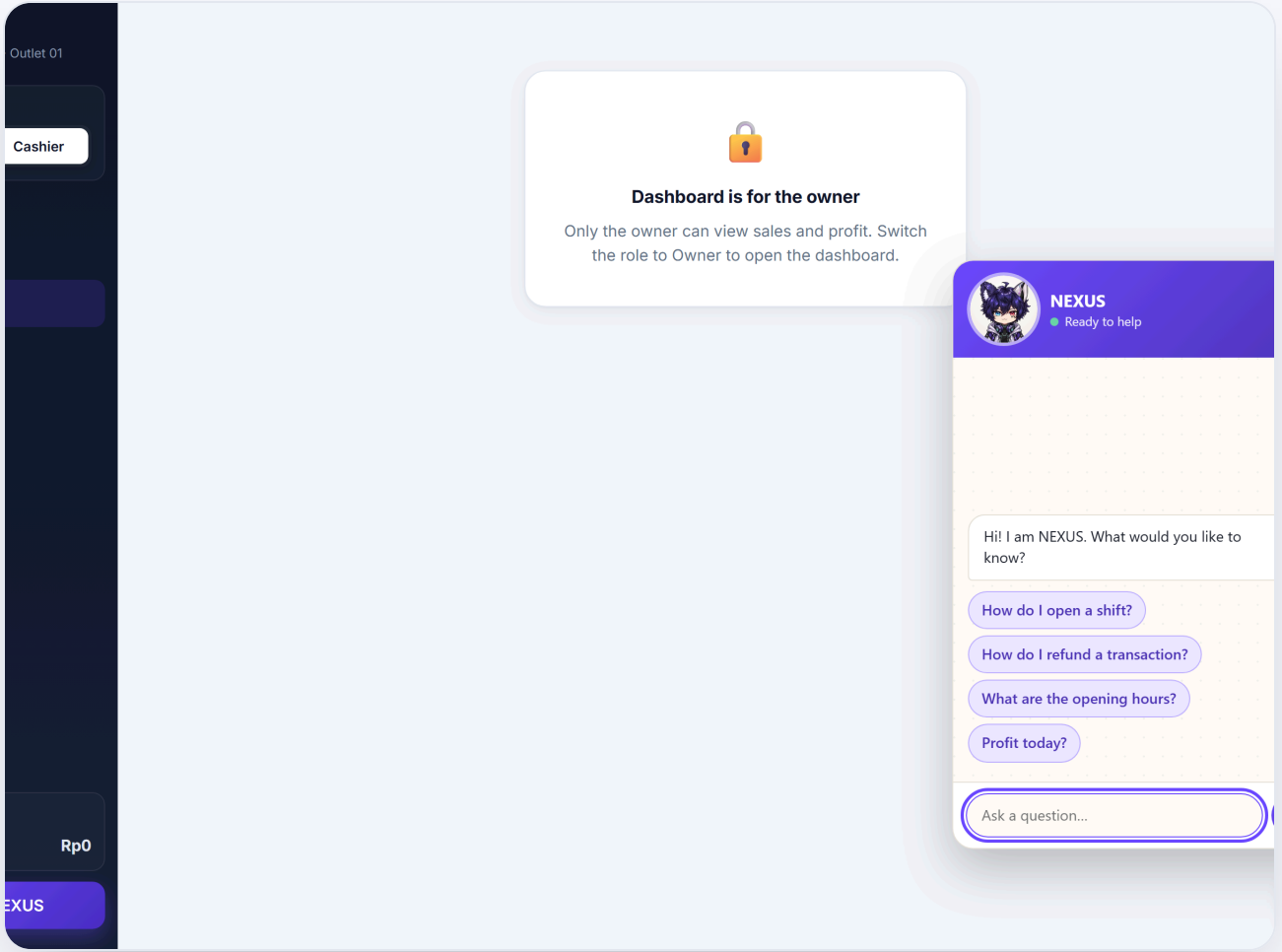


- **KPI cards** with change vs the previous period and margin.
- **Ranges:** Today, 7 days, 30 days or a custom date range.
- **Daily chart** of sales and profit, with a table view.
- **Top products** and **payment methods**.
- **Missing-COGS warning**, same rule NEXUS follows.

ROLE-BASED ACCESS

Cashiers can ask how, not how much

- **Dashboard locks** for the cashier role.
- **NEXUS refuses** a cashier's profit question: "Only the owner can view profit."
- **The refusal is audited** with `outcome=forbidden`.
- **SOP questions still work** for everyone.



ADMIN STUDIO

Admins run NEXUS without a developer

Knowledge

List, upload files, write FAQs, see processing status, version history with activate and rollback, soft delete.

Assistant

Name, persona prompt, language, LLM and fallback, score threshold, accent colour, allowed origins, avatar frames with live preview.

Insights

Knowledge gaps to resolve, thumbs-down answers, and an audit log filterable by action, outcome and date.

Secure sign-in

Supabase email login; only users listed as admins of that app get in.

Every change audited

Admin actions write the append-only audit log.

Same dark-sidebar look as the POS demo

SECURITY & GUARDRAILS

Safe by construction, not by prompt

Signed identity

RS256 JWT from the host, 10-minute TTL, verified per app via JWKS.

Tenant isolation

Retrieval always filters by app and active version; tools are registered per app.

Locked database

RLS on every table with no policies; only ai-service holds the service key.

Append-only audit

UPDATE and DELETE revoked and blocked by a trigger, even for the service role.

Abuse limits

CORS limited to allowed origins; 20 messages per minute per user.

No secrets in the browser

The widget holds no keys; Studio holds only the anon key.

MODELS & DATA

Swap the model per app; keep the data in one place

LLM gateway

- **Claude** adapter for cloud use.
- **OpenAI-compatible** adapter for self-hosted models (LM Studio, Ollama, vLLM). The demo runs qwen2.5-7b-instruct locally.
- **Fallback** adapter on failure, after one retry.
- **Fake** adapter for deterministic tests.

Supabase

- **Postgres + pgvector** with an HNSW index for chunks.
- **Storage** for knowledge files and avatar frames.
- **Conversations, feedback, gaps, audit** tables.
- **90-day** retention for conversations.

QUALITY

Tested at every layer

55

widget tests: states, corner snap, 401 retry, style isolation

29

Studio unit tests

E2E

Playwright: owner gets the exact profit, cashier refused, knowledge cited

≥90%

eval target; 100% on numbers and refusals

Unit

Date ranges in 3 time zones, rupiah parser, guardrail, role check, chunker.

Contract

mock-pos responses and SSE events checked against shared zod schemas.

Orchestrator

Scripted fake LLM: tool loop, guardrail replace, POS timeout, fallback.

NEXT STEPS

From pilot to production

Compliance

Decide external LLM use, data region, and retention for conversations, audit and knowledge gaps.

POS team

Agree the profit definition (refunds, discounts, VAT, missing COGS) and provide a staging endpoint.

Next tenants

Core Banking and ERP/HRIS reuse the same widget and service with their own knowledge and tools.



Thank you. Questions?